

Research Paper

Analysis of charging tariffs for residential electric vehicle users based on Stackelberg game

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ABSTRACT

Establishing a rational pricing mechanism for electric vehicles (EVs) can incentivize users to engage in charging activities, foster positive consumption behaviors, and indirectly benefit the grid by increasing load demand. Therefore, this paper conducts a Stackelberg game study between the grid and EV users by utilizing real charging and discharging data of EV users in F city. First, the price elasticity of EV users' electricity consumption behavior is analyzed. Second, the Stackelberg game model is constructed, multiple tariff schemes are designed to calculate the equilibrium tariff of the game, and the benefits of the grid and EV users are analyzed in comparison with the tariff policy that will be implemented in F city. Lastly, different seasonal scenarios are constructed to extend the applicability of the research results. It is found that Scheme 3 effectively incentivizes charging load growth, increases cross-subsidies and transmission and distribution fees, and protects the level of charging and discharging costs for residents; furthermore, lowering charging and discharging tariffs during seasons of high price elasticity for residents to increase demand is an optimal strategy for grid pricing.

1. Introduction

As the concepts of "carbon peak" and "carbon neutrality" gradually gain global consensus, attention to carbon emission reduction has significantly intensified. Consequently, green transportation has become the preferred choice for many. EVs, powered solely by electricity, emit zero carbon during operation, and their lifecycle emissions are steadily decreasing with the optimization of renewable energy structures (Iweh et al., 2021). Governments worldwide are promoting EV adoption, recognizing it as a pivotal technology to curb petroleum dependency and address climate change. In 2022, the Chinese government designated Hainan Province as the first pilot province for phasing out internal combustion engine vehicles, with plans to ban the sale of such vehicles province-wide by 2030. This initiative indicates an imminent surge in EV numbers. However, the unregulated integration of a growing number of EVs into regional power grids can significantly impact the quality of residential energy usage, the economic efficiency of grid operations, and reliability (Sharma et al., 2020; Torres et al., 2022). By scientifically managing the charging and discharging behaviors of a large number of EVs, the challenges posed by their uncontrolled integration into the grid can be addressed, thereby enhancing grid stability and leveraging the

energy storage capabilities of EVs (Srivastava et al., 2024). Thus, the primary focus of current research is on managing EV fleets from both technical and economic perspectives, ensuring rational dispatch of EV charging and discharging power.

1.1. Technical research aspects of EV dispatching

The technical research aspect mainly focuses on the analysis of EV usage characteristics and its interaction characteristics with the power system, based on which EV scheduling strategies are proposed. Mince Li et al. (2024) explored the current state of research on coupled modeling of EVs, transportation networks, and power grids, analyzed the optimization techniques related to power grid planning and EV charging/-discharging scheduling. In literatures (Jiang et al., 2024; Zhang et al., 2016; Jansen et al., 2024), the regulation potential and flexibility of EVs interacting with the power system have been analyzed, resulting in flexibility assessment parameters such as flexible capacity. Mohammad Hassan Bahmani et al. (2024), Markus Kreft et al. (2024) analyzed EV behavioral characteristics such as the number of EVs connected to the grid and charging duration by using non-stationary Markov chain and random forest prediction, respectively, which provided theoretical

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references for EV scheduling. Hengrong Zhang et al. (2024), Tianren Zhang et al. (2023) classified users based on EV behavioral characteristics to enhance the applicability of EV charging and discharging scheduling strategies. In literatures (Xie et al., 2024; Ireshika and Kepplinger, 2024), the uncertainty of EV charging and discharging demand is included in the considerations of scheduling strategies. The above scholars focus on the technical aspects of EV grid integration, flexibility, and behavioral characteristics to carry out EV scheduling research, which provides a mature theoretical reference for managing EV group.

1.2. Economic research aspects of EV dispatching

The economic research aspect mainly incentivizes EVs to adjust their charging and discharging behaviors or participate in dispatch management through EV charging and discharging prices, service strategies, and operation modes. In literatures (Chen et al., 2020; Lin et al., 2023a; Fu et al., 2023; Liu et al., 2023), non-cooperative game based on tariffs and loads is conducted between EVs and subjects such as distributed system operators or charging stations, and the tariff elasticity of loads is considered to formulate scheduling strategies; In literatures (Zare Gha-leh Seyyedi et al., 2022; Lin et al., 2023b), cooperative games have been used in the study of EVs scheduling strategies to improve the overall effectiveness of the game subjects; In literatures (Cao et al., 2022; Chen et al., 2023; Wang et al., 2023), the differentiated pricing strategy for EVs considers the interactions between the game strategies of each subject, and the objective equilibrium of each subject is achieved by the evolutionary game and Stackelberg game, respectively. In literatures (Straka et al., 2023; Hu et al., 2022), EV users were categorized based on their charging data, enabling charging service companies to develop differentiated power sales strategies or smart charging services based on various types of users. S. de la Torre et al. (2023), Wael Abdelfattah et al. (2024) proposed modes for EV participation in ancillary service markets or V2G strategies respectively to provide additional economic profit to EV users, thus incentivizing them to execute scheduling plans that are beneficial to system operation. The above scholars incentivize EV users to adjust their charging and discharging behaviors through tariffs, power sales strategies, and models that provide user revenues to promote the participation of EVs in system scheduling from an economic perspective, and the research results are expected to be practiced in the future.

1.3. Research summary

In summary, current research has extensively explored EV dispatching from both technical and economic perspectives. This research aims to achieve orderly participation of large-scale EVs in charging and discharging, thereby enhancing the security and reliability of grid operations.

However, the following research gaps persist: 1) a lack of consideration for seasonal variations in residents' price sensitivity; 2) insufficient analysis of the impact of EV load changes on grid transmission and distribution tariff revenues; and 3) the absence of comparisons between

study results and the effects of actual electricity pricing policies. Therefore, this paper proposes a study on the pricing strategy for charging and discharging (set by the grid) for EV users in City F. Given the information asymmetry between the grid and users, adjustments in charging and discharging behaviors are based solely on the grid's pricing, meaning users make decisions following the grid's actions. This context naturally aligns with a Stackelberg game scenario. Consequently, this paper develops a Stackelberg game model focusing on charging and discharging tariffs between the power grid and EV users, analyzing the benefits for both parties at the game equilibrium. The contributions and innovations of this study are as follows:

- Analyzing residential tariff elasticity across multiple seasonal scenarios by considering the sensitivity of EV users to tariffs in different seasons.
- On the grid side, in addition to considering the common target of electricity sales revenue, the study also introduces and analyzes the new indicator of transmission and distribution tariff revenue.
- Utilizing real electricity load data from City F and compared the results with the actual tariff policy to be implemented, aiming to provide a reference for tariff policy adjustments in City F.

The paper is structured as follows: Section 2 analyzes the price elasticity of residential electricity consumption behavior, Section 3 constructs a Stackelberg game model between the grid and EV users, Section 4 sets up three tariff scenarios for comparative analysis, Section 5 establishes different seasonal scenarios for analysis, and Section 6 concludes the study.

2. Grid-EV user transaction framework

In this study, the grid aims to maximize its revenue from electricity sales while implementing time-of-use pricing to regulate EV users' charging and discharging behaviors. Meanwhile, EV users subjectively adjust their charging and discharging behaviors based on time-of-use pricing, aiming to minimize their average charging costs over a certain period. Obviously, the grid's objective is to set as high a price as possible to maximize its revenue from electricity sales, which inevitably leads to an increase in the average charging costs for EV users. Based on this, it can be inferred that the grid and EV users have mutually exclusive objectives regarding the pricing of EV charging, thus leading to the construction of a game model and respective game strategies for both parties in the bargaining process.

Due to the nature of the game, the grid initiates the initial time-of-use pricing strategy to EV users, with the grid holding a dominant position in terms of information. Therefore, this game model falls under the category of leader-follower game. The game framework and strategies for the Grid-EV User interaction are illustrated in Fig. 1.

3. Analysis of price elasticity for EV users

The price elasticity matrix is derived from the traditional economic concept of the price elasticity of demand. It refers to the degree of

Table 1
Summary of key points in the literature.

Representative literature	Grid-connected analysis	Flexibility	Classification	EV characteristics	Electricity price gaming	Operating mode
(Li et al., 2024)	✓			✓		
(Jansen et al., 2024)		✓		✓		
(Kreft et al., 2024)				✓		
(Zhang and Qiu, 2024)	✓		✓			
(Lin et al., 2023a)	✓				✓	
(Lin et al., 2023b)				✓	✓	
(Wang et al., 2023)					✓	✓
(Hu et al., 2022)			✓			✓
(de la Torre et al., 2023)				✓		✓

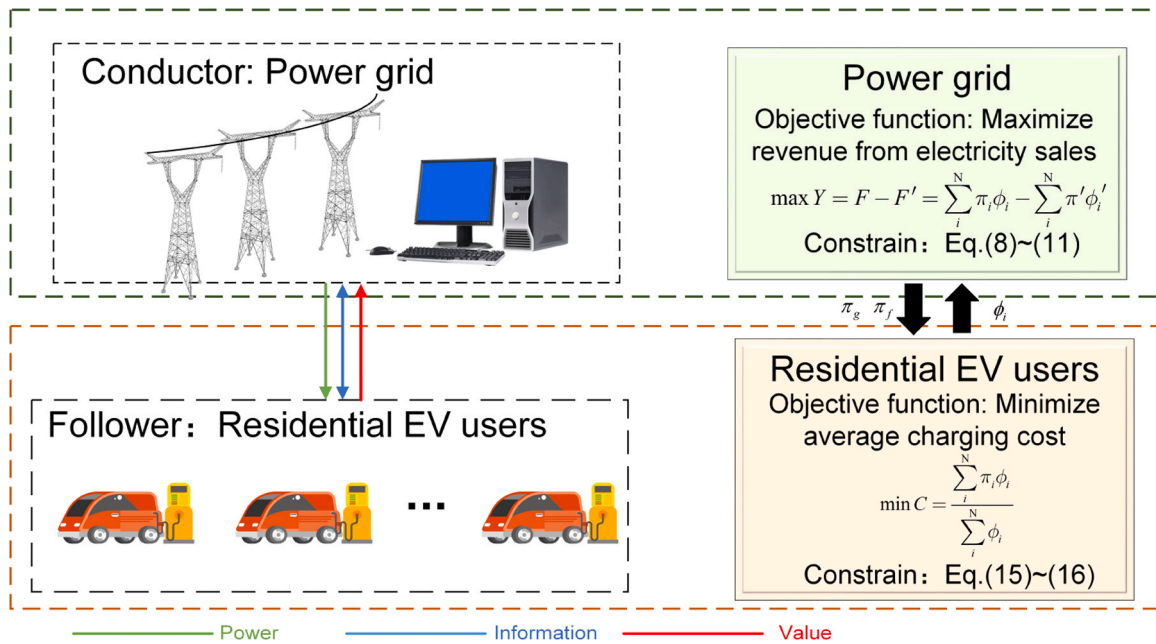


Fig. 1. Grid-Residential EV User Game Framework.

responsiveness of the quantity demanded of a commodity to changes in its price over a certain period. Specifically, it measures the percentage change in the quantity demanded of a commodity in response to a 1 % change in its price over a specific period (Greer, 2012). The price elasticity of demand is generally represented by the symbol (price elasticity of demand) and is calculated as shown in Eq. (1).

$$ED = \frac{\partial Q}{Q} \frac{P}{\partial P} \quad (1)$$

In the equation, Q represents the original quantity demanded of the commodity; ∂Q represents the change in demand resulting from the price response; P is the original price of the commodity; ∂P is the change in price of the commodity.

From Eq. (1), it can be observed that as the change in price ∂P decreases and the change in demand ∂Q increases, the price elasticity of demand ED becomes larger. This indicates that a slight change in the price of the commodity will result in a significant change in demand. Conversely, when the price elasticity of demand ED becomes smaller, a significant change in the price of the commodity will only lead to a slight change in demand. When ED equals 0, it signifies that regardless of the value of ∂P , ∂Q always remains 0, indicating complete insensitivity of the commodity to price changes, where demand remains unaffected by price fluctuations.

Electricity, as a commodity, exhibits a unique elasticity relationship between demand and price, differing from traditional goods. Purchasers respond to electricity prices in two ways: firstly, single-period response, where they react during specific periods of price changes, such as immediately shutting down certain electrical appliances when prices rise during that period. This response mechanism aligns closely with the principles of demand-price elasticity observed in traditional goods. Secondly, multi-period response involves adjusting load demands based on price fluctuations across different time periods, redistributing loads from high-price periods to low-price periods, for instance. However, even during single-period responses, users also consider past price fluctuations, evaluating whether adjusting loads during that period is worthwhile. Hence, the interaction between price and electricity consumption for users is not solely determined by the price of a single period.

Therefore, the concept of price elasticity matrix arises. Assuming one hour as a time interval, the residential electricity price elasticity

matrix E within a 24-hour scheduling cycle is formulated as follows (Greer, 2012):

$$E = \begin{bmatrix} \varepsilon_{1,1} & \varepsilon_{1,2} & \cdots & \varepsilon_{1,24} \\ \varepsilon_{2,1} & \varepsilon_{2,2} & \cdots & \varepsilon_{2,24} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{24,1} & \varepsilon_{24,2} & \cdots & \varepsilon_{24,24} \end{bmatrix} \quad (2)$$

In the equation, ε represents the price elasticity coefficient, where a higher value indicates a greater degree of change in electricity consumption in response to price variations during that time period. Furthermore, $\varepsilon_{i,i}$ and $\varepsilon_{i,j}$ are respectively referred to as the self-elasticity coefficient and the cross-elasticity coefficient. The self-elasticity coefficient $\varepsilon_{i,i}$ signifies the impact of changes in charging costs within time interval i the electricity demand within the same time interval; the cross-elasticity coefficient $\varepsilon_{i,j}$ denotes the responsiveness of electricity consumption changes within time interval i to variations in charging costs within time interval j . Their respective calculation methods are shown in Eqs. (3) and (4).

$$\varepsilon_{i,i} = \frac{\partial Q_i}{Q_i} \frac{P_i}{\partial P_i} \quad (3)$$

$$\varepsilon_{i,j} = \frac{\partial Q_i}{Q_i} \frac{P_j}{\partial P_j} \quad (4)$$

In the equation, subscripts i and j respectively denote the time interval indices.

The calculation method for the price elasticity matrix is illustrated in Eq. (5).

$$\begin{bmatrix} \frac{\partial Q_1}{Q_1} & \frac{\partial Q_2}{Q_2} & \cdots & \frac{\partial Q_{24}}{Q_{24}} \end{bmatrix}^T = E \begin{bmatrix} \frac{\partial P_1}{P_1} & \frac{\partial P_2}{P_2} & \cdots & \frac{\partial P_{24}}{P_{24}} \end{bmatrix}^T \quad (5)$$

4. Mutual benefit models for both the grid and EV users

4.1. Grid-Side Model

The grid-side model involves the formulation of time-of-use pricing strategies for residential EV users to encourage adjustments in their charging behaviors, thereby assisting the grid in peak shaving and load shifting, i.e., transferring high peak loads to low-demand periods.

Residential users consider various factors when charging their EVs, aiming to minimize their electricity costs. If the implementation of time-of-use pricing results in higher electricity expenses for users compared to before, residents may opt for alternative charging methods, leading to a loss of revenue for the grid from residential EV charging fees and a decrease in overall societal benefits. Therefore, the total expenditure for users should not increase after the implementation of time-of-use pricing. Assuming the total expenditure by users before implementing time-of-use pricing is F , and after implementing time-of-use pricing is F' , then:

$$F' = \sum_i^N \pi'_i \phi'_i \quad (6)$$

$$F = \sum_i^N \pi_i \phi_i \quad (7)$$

In the equation, π' represents the original charging price for residential EVs; π_i represents the charging price for residential EVs during the respective time interval after the implementation of time-of-use pricing policy; ϕ'_i represents the amount of charging for residential EVs during the respective time interval; ϕ_i represents the amount of charging for residential EVs during the respective time interval after the implementation of time-of-use pricing policy.

Under peak-valley time-of-use pricing incentives, residential users, aiming to reduce their expenditure on charging their EVs, shift a portion of their charging load from peak periods to valley periods. Therefore, the post-implementation expenditure should not exceed the original expenditure, as shown in Eq. (8).

$$F \leq F' \quad (8)$$

The aforementioned model imposes constraints on the demand response effect after the implementation of time-of-use pricing. However, the reasonable setting of peak and valley electricity prices in time-of-use pricing can further support demand response and achieve a "win-win" situation for both the grid and residential users. The purpose of implementing peak-valley time-of-use pricing is to incentivize users to charge their EVs more during valley and shoulder periods, reducing peak electricity consumption and thereby achieving peak shaving and load balancing objectives to enhance grid efficiency. Therefore, peak electricity prices are set higher than shoulder prices, and shoulder prices are set higher than valley prices, as shown in Eq. (9).

$$\pi_g \leq \pi_p \leq \pi_f \quad (9)$$

In the equation, π_p represents the electricity price for residential EVs during shoulder periods, typically set at the level before the implementation of time-of-use pricing π' .

Additionally, when establishing peak and valley electricity prices, the fluctuation ratio between peak and valley prices needs to be controlled within a reasonable range to avoid phenomena such as inversion of charging load peaks and valleys or inadequate user response, which may prevent the smooth shaping of the load curve. Therefore, it is required that the fluctuation ratio between peak and valley electricity prices remains within a certain range, as shown in Eqs. (10) and (11).

$$1.1 \leq \frac{\pi_f}{\pi_p} \leq 1.5 \quad (10)$$

$$0.5 \leq \frac{\pi_g}{\pi_p} \leq 0.9 \quad (11)$$

To ensure that the implementation of time-of-use pricing policy benefits the grid in terms of charging and discharging revenue recovery, the change in grid electricity sales revenue before and after the implementation of time-of-use pricing is considered as the objective function, as depicted in Eq. (12).

$$\max Y = F - F' = \sum_i^N \pi_i \phi_i - \sum_i^N \pi'_i \phi'_i \quad (12)$$

By implementing time-of-use pricing, the distribution of residential EV charging and discharging loads is effectively improved. Using this optimized load distribution as a reference, the impact of two schemes—original electricity pricing and time-of-use pricing—on grid benefits during peak and valley periods is examined. The calculation methods are illustrated in Eqs. (13) and (14).

$$Y_f = \pi_f \phi_f - \pi'_f \phi'_f \quad (13)$$

$$Y_g = \pi_g \phi_g - \pi'_g \phi'_g \quad (14)$$

In the equations, Y_f and Y_g respectively represent the changes in grid benefits during peak and valley periods.

4.2. EV user model

The charging and discharging behavior of EV users exhibit a notable sensitivity to charging tariffs. This implies that the charging electricity consumption of users throughout various time periods across the province is influenced by the prevailing time-of-use electricity pricing levels. Consequently, the formulation of time-of-use tariffs enables the expression of the expected charging and discharging volumes for EVs across different time segments (Gaurav Kansal and Rajive Tiwari, 2023), as represented by Eq. (15).

$$\begin{bmatrix} \bar{\phi}_1 \\ \bar{\phi}_2 \\ \vdots \\ \bar{\phi}_{24} \end{bmatrix} = \begin{bmatrix} \phi'_1 & 0 & \dots & 0 \\ 0 & \phi'_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \phi'_{24} \end{bmatrix} \mathbf{E} \begin{bmatrix} \partial \pi_1 / \pi' \\ \partial \pi_2 / \pi' \\ \vdots \\ \partial \pi_{24} / \pi' \end{bmatrix} + \begin{bmatrix} \phi'_1 \\ \phi'_2 \\ \vdots \\ \phi'_{24} \end{bmatrix} \quad (15)$$

In the equation, $\bar{\phi}_i$ represents the expected charging quantity of residential EVs following the implementation of time-of-use electricity pricing by the grid; $\partial \pi_i$ represents the variation in time-of-use electricity pricing; $\mathbf{E}$ represents the elasticity matrix of residential electricity pricing.

Following the implementation of time-of-use pricing, in order to prevent phenomena such as "peak-valley inversion" or the concentration of all residents charging during periods of low electricity prices, the actual charging quantity of residential EV users should fluctuate within the allowable range of the grid's expected charging quantity, that is:

$$\left| \frac{\phi_i - \bar{\phi}_i}{\bar{\phi}_i} \right| \leq \alpha \quad (16)$$

In the equation, α represents a known parameter, which constrains the range of fluctuation for the actual charging quantity of residential EVs following scheduling.

Residential EV users adjust their charging and discharging behavior based on time-of-use pricing policies, with their primary objective being to minimize their average charging cost. This is achieved by minimizing the ratio of charging costs to total charging quantity within the scheduling period, as demonstrated by Eq. (17).

$$\min C = \frac{\sum_i^N \pi_i \phi_i}{\sum_i^N \phi_i} \quad (17)$$

In the equation, C represents the average charging cost for residential EVs.

4.3. Demand response assessment model

The effectiveness of demand response achieved by time-of-use pricing policies is primarily assessed from two perspectives: social benefits and system security operation. Among these, assuming the unit cost for

the power grid to purchase electricity from generating enterprises in their time period is π_i^e , Eq. (18) demonstrate the calculation method for the social benefit index (Wu et al., 2024).

$$B = \sum_i^N (\pi_i \phi_i - \pi_i^e \phi_i) \quad (18)$$

In the equation, B represents the social benefits achieved after the implementation of time-of-use pricing.

The assessment of system security operation primarily focuses on

two indicators: peak-to-valley difference and load ratio. The peak-to-valley difference reflects the extent of peak shaving and valley filling required by a regional power grid, and the analysis of peak-to-valley difference serves as the basis for formulating peak shaving and valley filling schemes, as well as load management. The load ratio indicates the balance of EV charging and discharging and serves as a measure of load fluctuation within a day. A higher daily load ratio indicates a more balanced load fluctuation. Balanced electricity consumption is a prerequisite for increasing the load ratio and maintaining regional elec-

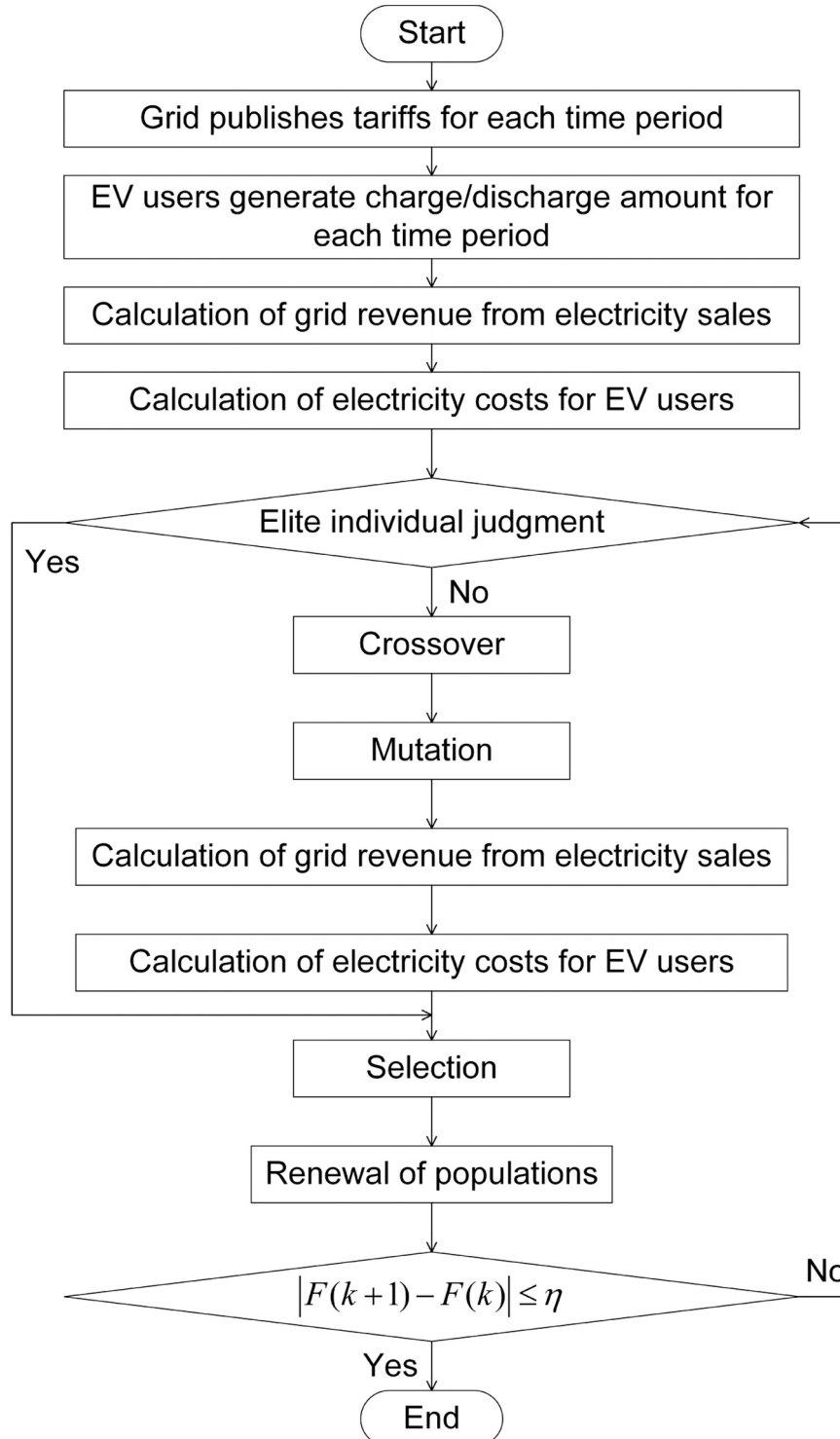


Fig. 2. NSGA-II solving process.

tricity balance, which is highly beneficial for the reliable operation of the power grid. The calculation methods for these two indicators are demonstrated by Eqs. (19) and (20).

$$FG = \phi_{\max} - \phi_{\min} \quad (19)$$

$$LF = \frac{\sum_{i=1}^N \phi_i / 24}{\phi_{\max}} \quad (20)$$

In the equation, FG represents the peak-to-valley difference; $\phi_{\max}$, $\phi_{\min}$ respectively represent the maximum and minimum load of residential EVs within the scheduling period; LF represents the load ratio.

5. Algorithmic process

The current research on the Stackelberg model is quite mature, with most studies using genetic algorithms for solutions. This paper aims to explore the impacts of the EV tariff mechanism on the grid and users, providing a reference for adjusting real-world tariff policies, making algorithm innovation not the focus. Therefore, the improved genetic algorithm (NSGA-II), known for its strong scalability and search capability, is directly used to solve the Stackelberg game between the grid and EV users. The solution process is detailed below.

6. Example analysis

6.1. Example parameters

Currently, residential EV users in a certain province are still charged for charging based on a combined-meter user electricity rate. Therefore, the parameter for the residential electricity rate in City F, set at 0.533 CNY/kWh, is adopted for analysis, denoted as $\pi = 0.533$ CNY/kWh and $\pi_p = 0.533$ CNY/kWh.

Based on an analysis of the charging and discharging characteristics of 227 residential EVs in City F and a K-means clustering study, combined with the number of personal charging stations for residents in City F in 2023, the load profile for residential EV charging and discharging periods in the province is fitted, as depicted in Fig. 3.

To effectively evaluate the superiority of the proposed EV time-of-use pricing scheme, three types of schemes are designed to compare the variations in grid revenue and average residential electricity costs. The settings for each scheme are as follows:

Scheme 1: peak hours : 11 : 00–14 : 00、18 : 00–23 : 00; shoulder hours : 7 : 00–11 : 00、14 : 00–18 : 00 ; Valley hours: 23:00-next day

7:00.

Scheme 2: peak hours: 8:00–22:00; valley hours: 22:00 to the next day at 8:00.

Scheme 3: peak hours: 8:00–22:00; valley hours: 22:00 to the next day at 8:00. Peak electricity price: 0.5630 CNY/kWh, valley electricity price: 0.3330 CNY/kWh (the actual tariff policy to be implemented in City F).

6.2. Analysis of Scheme 1

Based on the residential electricity consumption load data in City F and the calculated residential electricity pricing elasticity matrix, considering the use of the current summer elasticity matrix, and combined with the time period division of peak, shoulder, and valley hours in this case study, the elasticity matrix for electricity pricing is adjusted. The final electricity pricing elasticity matrix is presented in Table 2.

Based on the model constructed in Section 3 and the parameters input in Section 4.1, a Stackelberg game model is established. The iteration is set to 20 times, yielding the time-of-use pricing scheme and the iteration process as illustrated in the Fig. 4 and Table 3.

From the above figure, it can be observed that the model exhibits good convergence, with convergence achieved after the fourth iteration. It is noted that the peak-hour electricity price consistently adopts the maximum value within the model-defined range, representing a 50 % increase from the shoulder-hour electricity price of 0.533 CNY. This increase suggests that raising the peak-hour electricity price contributes to the realization of grid revenue and demand response effectiveness. Conversely, the valley-hour electricity price ultimately computes to 0.4493 CNY, reflecting a decrease of 15.70 % compared to the flat-rate electricity price.

Following the implementation of time-of-use pricing, the objective functions for both the grid side (change in electricity sales revenue) and the residential EV side (average charging cost) are presented in Table 4.

From the above table, it is evident that after the implementation of time-of-use pricing, the grid-side electricity sales revenue is zero, indicating the absence of grid losses. Meanwhile, the average charging tariff

Table 2
Example of Electricity Price Elasticity Matrix.

	Valley	Shoulder	Peak
Valley	−0.1	0.01	0.012
Shoulder	0.01	−0.1	0.016
Peak	0.012	0.016	−0.1

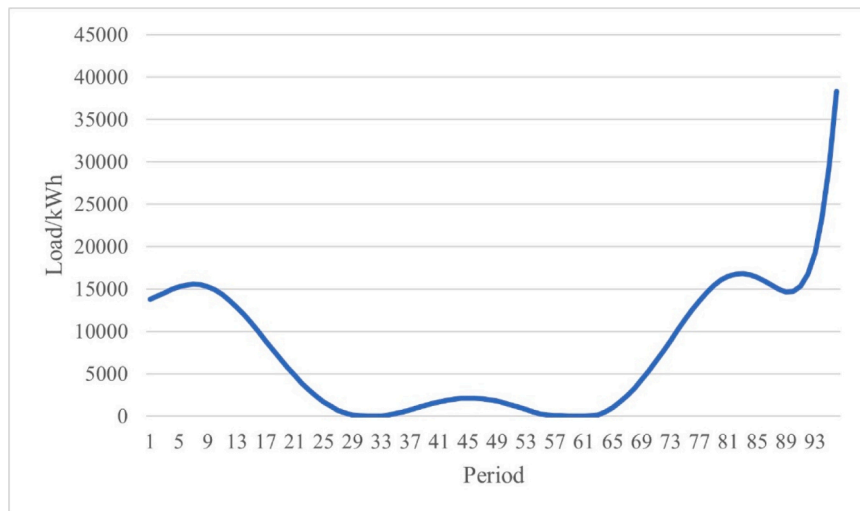


Fig. 3. Load Curve of Residential EV Charging and Discharging in City F.

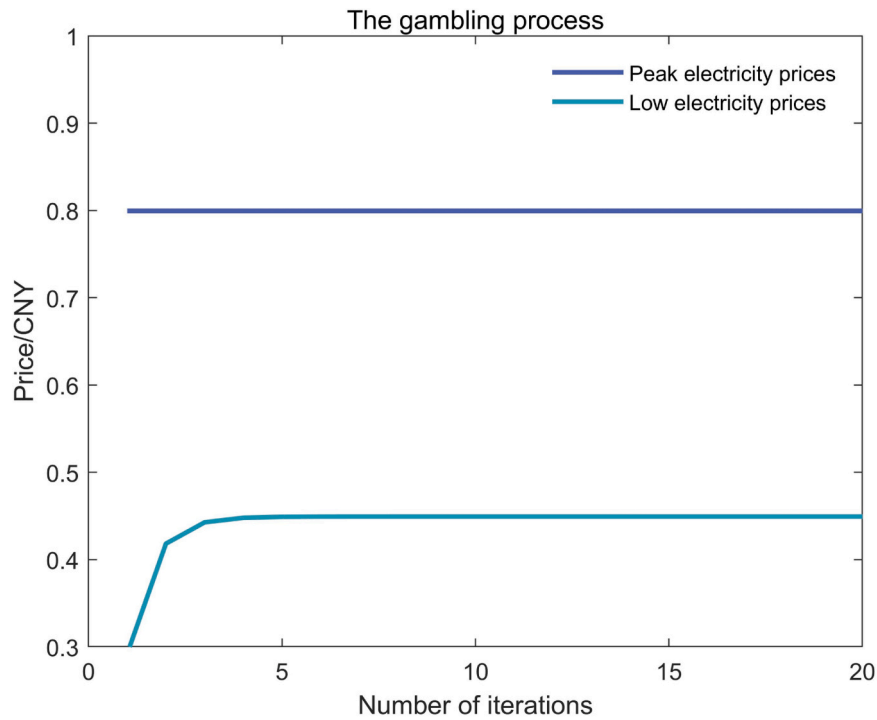


Fig. 4. Iterative Process of Peak-Valley Pricing Game.

Table 3
Calculated Results of Time-of-Use Pricing Scheme.

Time Period Division	Shoulder Period	Valley Period,	Peak Period
Charging Tariff/(CNY /kWh)	0.5330	0.4493	0.7995

Table 4
Benefits for Grid and Residential EVs under Time-of-Use Pricing Scheme.

Objectives of Each Entity	Value
Grid Side	0
Residential EV Side	0.4709

for residential EVs stands at 0.4709 CNY, representing a decrease of 0.0621 CNY compared to the pre-time-of-use pricing rate of 0.533 CNY.

Further investigation into the load profile of residential EVs after the implementation of time-of-use pricing is illustrated in Fig. 5.

As depicted in Fig. 5, the implementation of time-of-use pricing has led to a reduction in the charging load of EVs during the late peak period from 18:00–23:00. The peak charging load occurs at 21:00, amounting to 48,577.03 kWh, representing a decrease of 27.35 % compared to the pre-time-of-use pricing period. Concurrently, there is an increase in the load of EVs during the low valley period from 24:00–5:00 the next day. The maximum load during this period, at 24:00, reaches 120,680.64 kWh, indicating a rise of 9.05 % compared to the pre-time-of-use pricing period. Additionally, there is a decrease in the load of EVs during the noon peak period from 11:00–14:00.

Moreover, prior to the implementation of time-of-use pricing, the total load demand of residential EVs in City F was 737,709.75 kWh per day. After the implementation of time-of-use pricing, the total load demand decreased to 678,443.09 kWh, reflecting an 8.03 % reduction in load demand. This reduction in demand essentially preserves the psychological consumption pattern of EV owners.

In summary, the implementation of time-of-use pricing demonstrates effective peak shaving and valley filling capabilities, guiding residents to

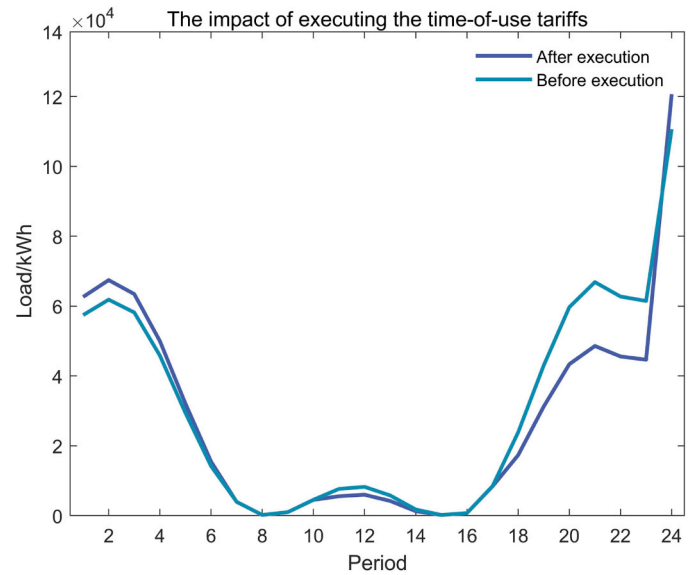


Fig. 5. Variation in charging and discharging load of residential EVs in city F.

adjust their EV charging behavior and reducing the average cost of charging for users. This underscores its significant practical value.

6.3. Analysis of Scheme 2

In this section, the elasticity matrix for electricity pricing utilizes the

Table 5
Residential Electricity Price Elasticity Matrix for City F (Summer).

	Valley Period	Peak Period
Valley Period	−0.1112	0.0830
Peak Period	0.0830	−0.1112

summer elasticity matrix for residents of City F, as presented in Table 5.

The results of the Stackelberg game model for this scheme are presented in Fig. 6 and Table 6.

From the above figure, it can be observed that the model exhibits good convergence, with convergence achieved after the second iteration. Similarly, the peak-hour electricity price also adopts the maximum value within the model-defined range. The valley-hour electricity price ultimately computes to 0.4797 CNY, representing a decrease of 10.00 % compared to the shoulder-hour electricity price.

Following the implementation of time-of-use pricing, the objective functions for both the grid side (change in electricity sales revenue) and the residential EV side (average charging cost) are presented in Table 7.

The above table indicates that following the implementation of time-of-use pricing, the grid-side revenue has increased by 15260.76 CNY compared to before the implementation of time-of-use pricing, resulting in a slight increase in electricity sales revenue. The average charging tariff for residential EVs stands at 0.5050 CNY, representing a decrease of 0.0250 CNY compared to the pre-time-of-use pricing rate of 0.533 CNY.

Further investigation into the load profile of residential EVs after the implementation of time-of-use pricing is illustrated in Fig. 7.

From Fig. 7, it is evident that the implementation of time-of-use pricing has led to a significant reduction in the charging load of EVs during the late peak periods from 18:00–23:00 and during the noon peak periods from 11:00–14:00. This reduction is primarily attributed to the absence of a shoulder period for residents to shift their loads. Consequently, users can only transfer electricity from peak to valley periods, and the significant difference in peak and valley electricity prices leads to higher elasticity coefficients in the residential electricity price elasticity matrix. Additionally, there is a substantial increase in the load of EVs during the low valley period from 24:00–5:00 the next day. The maximum load during this period, at 24:00, amounts to 186,022.60 kWh, representing a 68.10 % increase compared to the pre-time-of-use pricing period. This effectively encourages residents to charge their EVs during the late-night valley periods.

However, the implementation of this scheme has resulted in a total daily charging and discharging demand of 808,880.17 kWh for residential EVs, leading to a 9.65 % increase in residential charging and discharging load demand within a day, thereby promoting positive consumption trends.

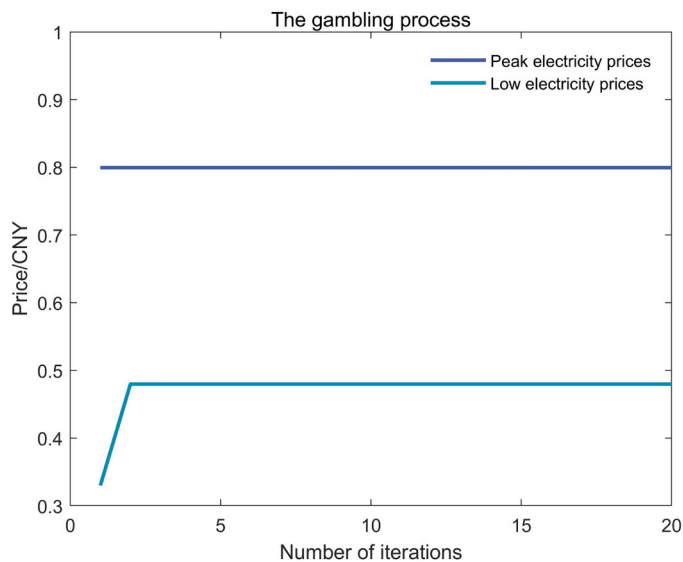


Fig. 6. Iterative Process of Peak-Valley Pricing Game.

Table 6
Calculated Results of Time-of-Use Pricing Scheme.

Time Period Division	Valley Period	Peak Period
Charging Tariff/(CNY/kWh)	0.4797	0.7995

Table 7
Benefits for Grid and Residential EVs under Time-of-Use Pricing Scheme.

Each Entity's Objectives	Values
Grid Side	15260.76
Residential EV Side	0.5050

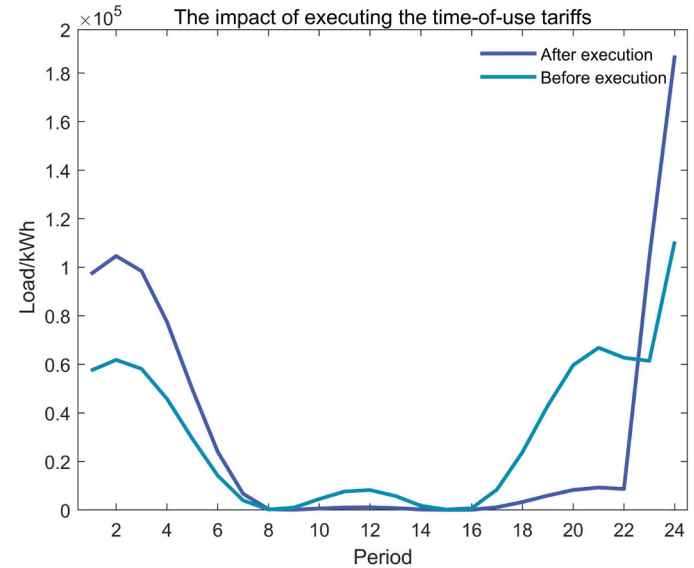


Fig. 7. Variation in Charging and Discharging Load of Residential EVs in City F.

6.4. Comparative analysis of scheme benefits

Based on the results of the tariff calculations for Schemes 1 and 2, the specifics of the three schemes are as follows:

Scheme 1: Peak Hours: 11:00–14:00, 18:00–23:00; Shoulder Hours: 7:00–11:00, 14:00–18:00; Valley Hours: 23:00–next day 7:00. Peak electricity price: 0.7995 CNY/kWh, Valley electricity price: 0.5330 CNY/kWh, Valley electricity price: 0.4493 CNY/kWh.

Scheme 2: Peak Hours: 8:00–22:00; Valley Hours: 22:00–next day 8:00. Peak electricity price: 0.7995 CNY/kWh, Valley electricity price: 0.4797 CNY/kWh.

Scheme 3: Peak Hours: 8:00–22:00; Valley Hours: 22:00–next day 8:00. Peak electricity price: 0.5630 CNY/kWh, Valley electricity price: 0.3330 CNY/kWh.

The analysis of the benefits of the three schemes is presented in

Table 8
Comparative Analysis of Multi-Scheme Benefits.

Scheme	Grid-Side Benefits (CNY)	Peak Period Benefit Variation (CNY)	Valley Period Benefit Variation (CNY)	Residential-Side Benefits (CNY)
Scheme 0	0	59931.49	−34789.95	0.4709
Scheme 1	15260.76	17033.53	−39706.60	0.5050
Scheme 2	−74744.78	5309.84	−131413.70	0.3818
Scheme 3				

Table 8.

Following the implementation of Schemes 1 and 2, the EV charging and discharging load of residents in City F is illustrated in Fig. 5 and Fig. 7, respectively. The EV charging and discharging load for City F residents under Scheme 3 is depicted in Fig. 8.

By examining Table 8 in conjunction with Fig. 8, it is observed that Scheme 3 incurs significant losses for the grid, with a reduction in electricity sales revenue of 74,744.78 CNY compared to the pre-implementation period. This is primarily due to the excessively low valley electricity prices under this scheme, which incentivize users to increase their charging and discharging load during valley hours. As a result, the grid bears the burden of a substantial amount of low-priced load consumption during these hours, leading to considerable losses in grid revenue. Additionally, the marginal increase in peak electricity prices coupled with a significant decrease in electricity consumption exacerbates the overall loss. Furthermore, Scheme 3 exhibits the lowest average residential electricity cost among the three schemes, at 0.3818 CNY/kWh, indicating favorable benefits for residents with EVs. Moreover, the implementation of time-of-use pricing under Scheme 3 results in a total demand for residential charging and discharging of 3371,738.02 kWh, representing a 9.83 % increase compared to the non-time-of-use pricing scenario, effectively incentivizing residential electricity consumption.

Further calculations are conducted to determine the scale of cross-subsidies for residents in City F under the three schemes, based on the total electricity consumption for EV charging and discharging for July 2023 and 2022 across the entire province. The benefits of the three schemes are summarized in Table 9.

Upon comprehensive analysis from the following perspectives, Scheme 3 emerges as the optimal choice overall:

(1) Impact on Charging Volume: Following the implementation of Scheme 3, the total residential charging demand increases to 3371,738.02 kWh, representing a 9.83 % increase compared to the non-time-of-use policy. This increase is notably higher than the respective 17 % and 7.84 % increases observed in Scheme 1 and Scheme 2, indicating a more proactive positive consumption promotion effect.

(2) Impact on Time-of-Use Revenues/Costs: Under the current residential EV charging and discharging demand of Scheme 3, the monthly scale of residential electricity price cross-subsidies is projected to increase by 36.75 million CNY. When scaled to the 2022 electricity consumption level, this increase can reach up to 248 million CNY, surpassing the respective increases of 248 million CNY and 191 million

CNY in Scheme 1 and Scheme 2.

(3) Impact on Transmission and Distribution Fees: Scheme 3 exhibits the highest demand response incentive effect. According to calculations based on the 2022 scale of EV charging and discharging, Scheme 3 can positively increase transmission and distribution fees by 50.58 million CNY. This increase exceeds the respective increments of 87.47 million CNY and 40.34 million CNY in Scheme 1 and Scheme 2, thereby contributing the most to the operational efficiency of the grid.

(4) Impact on Residential Charging Benefits: Under Scheme 3, the average residential electricity cost is 0.3818 CNY/kWh, representing a decrease of 0.0891 CNY/kWh and 0.1232 CNY/kWh compared to Scheme 1 and Scheme 2, respectively. Hence, the cost of residential charging is minimized, resulting in the highest electricity usage benefits.

6.5. Sensitivity analysis

The load of EV users under different charging and discharging prices is influenced by their sensitivity to electricity prices, which in turn affects grid revenue from electricity sales and the average cost of electricity for users. Therefore, this section introduces the sensitivity coefficient β to adjust the EV user price elasticity matrix E , i.e. $E' = \beta E$. The changes in benefits for the grid and users under different sensitivity coefficients are shown in Fig. 9.

In the sensitivity analysis process, the valley-time electricity price of Scheme 1 increases with the increase of electricity price elasticity: when β rises from 0.7 to 1.3, the valley-time electricity price is 0.3982, 0.4145, 0.4315, 0.4493, 0.4684, 0.4797, 0.4797 CNY/kWh, respectively. The electricity price of Scheme 2 doesn't change with the change of electricity price elasticity. Combined with Fig. 9, it can be seen that Scheme 1 ensures that the grid benefits remain unchanged from $\beta = 0.7$ to $\beta = 1.1$ by increasing the valley time tariff, and the users also reduce the average cost of electricity by reducing the load and adjusting the behavior of electricity consumption. In Schemes 2 and 3, since the tariff does not change, the grid and user benefits are mainly affected by the growth of user load. It is worth noting that when $\beta = 1.2$ in Scheme 2, the residential user benefit changes from a downward trend to a stable trend, which is due to the fact that Scheme 2 only sets the peak-time tariff (0.7995 CNY/kWh) and the valley-time tariff (0.4797 CNY/kWh), and as the tariff elasticity rises, the users end up using electricity only in the valley time, which results in the average cost of electricity consumption remaining unchanged. Meanwhile, the load of electricity consumption continues to rise, the benefits from the size gap are greater than those from the price gap, leading to a shift from a downward trend to an upward trend in grid benefits.

7. Analysis considering seasonal scenarios

7.1. Example parameters

Given the close correlation between residential electricity usage patterns and the elasticity of quantity and price with seasons, this study gathers comprehensive electricity consumption data from residents in City F for April 2022, July 2022, October 2022, and January 2023. These months represent the spring, summer, autumn, and winter seasons, respectively. The data includes the full volume of electricity consumption for residents using peak and valley pricing schemes. The sample size is approximately 800,000. By aggregating peak and valley electricity consumption data for residents in City F across various months, the elasticity relationship between electricity consumption and prices is calculated based on Eq. (1). Subsequently, time-of-use pricing schemes are established for multiple seasonal scenarios, and the impact of each scheme on resident charging volume, charging costs, and grid profit and loss is analyzed. This theoretical analysis provides support for the proposed seasonal time-of-use pricing strategy for EV users in City F.

Since a detailed analysis of the typical summer scenario was conducted in Section 3, this section focuses on analyzing time-of-use pricing

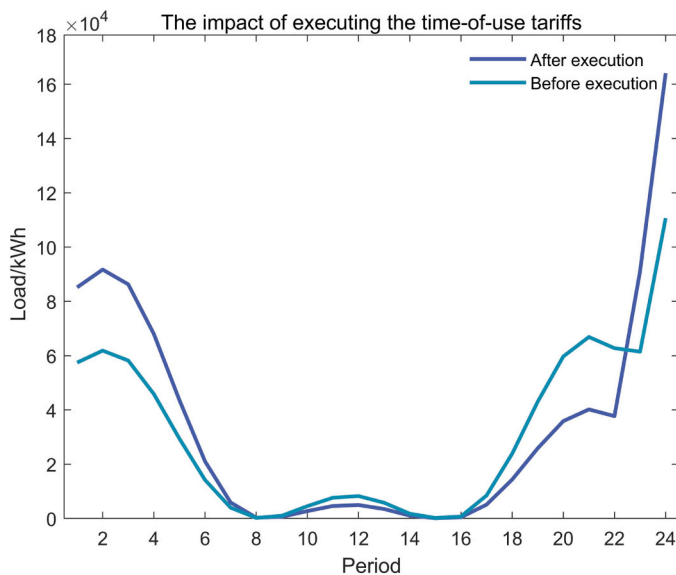


Fig. 8. Variations in EV Charging and Discharging Load for Residents in City F.

Table 9
Comparative Analysis of Multi-Scheme Benefits for City F in July 2023 and 2022 Under Time-of-Use Pricing.

Scheme	July 2023(CNY)		Year 2022(CNY)	
	Time-of-Use Profit/Loss	Transmission and Distribution Fees	Time-of-Use Profit/Loss	Transmission and Distribution Fees
Scheme 1	0.00	−5465575.33	0.00	−36889275.64
Scheme 2	−8479468.88	1517008.12	−57231205.49	10238872.84
Scheme 3	−36753812.92	7494202.75	−248065657.05	50581264.38

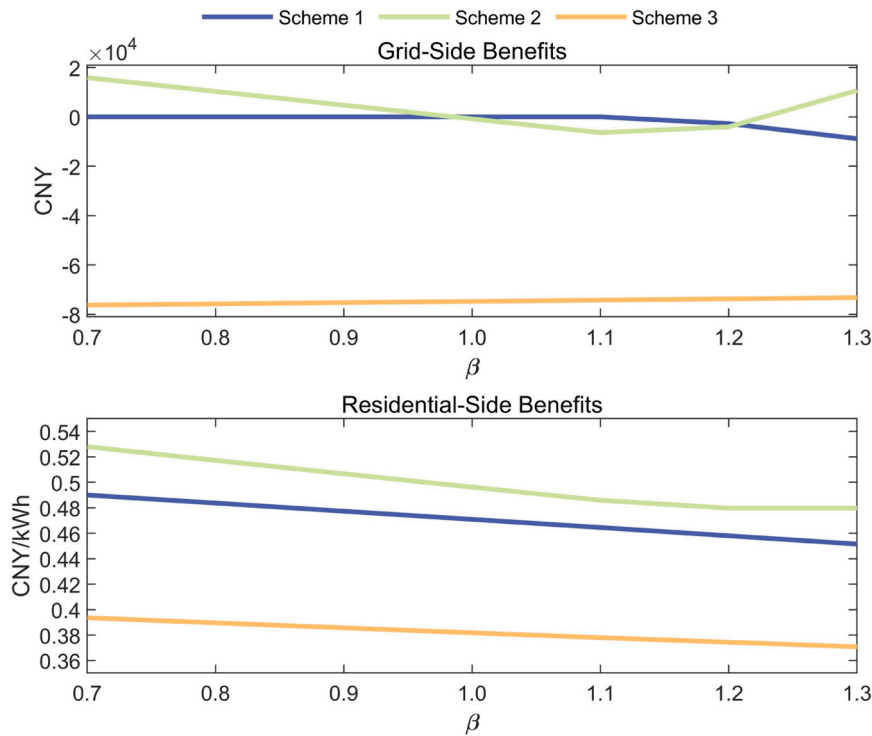


Fig. 9. Sensitivity analysis of benefits for grid, residential EV users.

schemes for the typical spring, autumn, and winter seasons. Peak hours are set from 8:00–22:00, while valley hours are set from 22:00–8:00 the following day. The elasticity matrices of electricity prices for each season are presented in Table 10.

7.2. Analysis of results

Building upon the preceding analysis of various time-of-use pricing schemes for EV users in a certain province, and in conjunction with the existing time-of-use pricing structure in the province, this section integrates and compares the time-of-use pricing strategies and the benefits to the grid and users obtained from three typical seasonal scenarios. These are contrasted with the relevant results derived from the summer scenario analysis presented in Section 3. The comparative outcomes are

Table 10
Elasticity Matrices of Electricity Prices for Residential Spring, Autumn and Winter Seasons in City F.

Spring	Valley Period	Peak Period
Valley Period	−0.4681	0.3495
Peak Period	0.3495	−0.4681
Autumn	Valley	Peak
Valley Period	−0.8223	0.6014
Peak Period	0.6014	−0.8223
Winter	Valley	Peak
Valley Period	−1.1595	0.8658
Peak Period	0.8658	−1.1595

illustrated in Fig. 10 and summarized in Table 11.

From Fig. 10 and Table 11, it is evident that due to the higher elasticity of residential electricity prices in spring, autumn, and winter compared to summer, the grid tends to lower selling prices to boost demand. Specifically, during spring and autumn, there is a slight

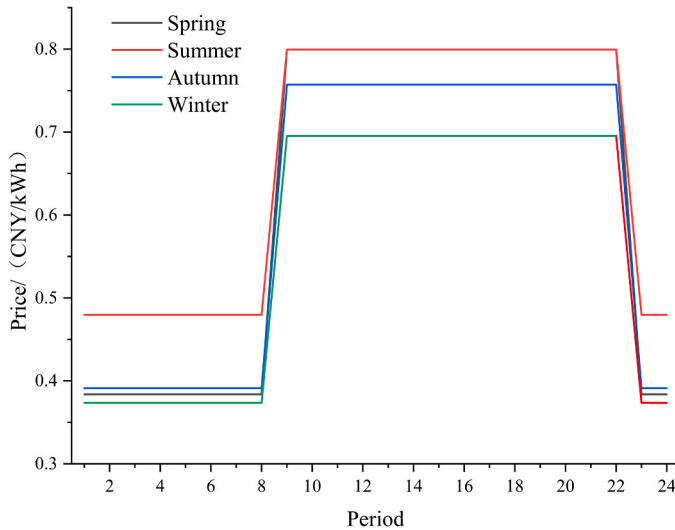


Fig. 10. Time-of-Use Pricing for Residential EVs in Typical Seasons.

Table 11
Game Results of Grid-Residential EV Time-of-Use Pricing in Typical Seasons.

Scenario	Peak Electricity Price (CNY/kWh)	Valley Electricity Price (CNY/kWh)	Grid-Side Benefits (CNY)	User Benefits (CNY/kWh)	Load Demand (kWh)
Spring	0.7995	0.3838	−150295.22	0.4627	3211416.81
Summer	0.7995	0.4797	−75447.74	0.4985	3131089.77
Autumn	0.7570	0.3911	−277186.15	0.4211	3227733.79
Winter	0.6954	0.3734	−277938.12	0.4035	3365755.00

reduction in valley electricity prices compared to summer, with peak electricity prices remaining relatively stable in spring and decreasing by 0.0425 CNY/kWh in autumn. In winter, both peak and valley electricity prices are significantly reduced compared to summer, with peak prices decreasing by 10.79 % and valley prices by 22.16 %. This reduction is attributed to the highest elasticity of residential electricity consumption in winter, where price changes have a greater impact on load demand. This is evident from the maximum load demand reached during winter, totaling 3365,755.00 kWh.

Simultaneously, the average charging and discharging costs for residential EVs under time-of-use pricing schemes in spring, autumn, and winter are lower compared to summer. The most notable reduction is observed in winter, with user benefits amounting to 0.4035 CNY/kWh, representing a decrease of 19.06 % compared to summer. However, on the grid side, implementing time-of-use pricing results in a significant increase in revenue loss due to the overall decline in peak and valley electricity prices.

The charging load for residential EVs following the implementation of time-of-use pricing schemes in each season is illustrated in Fig. 11.

From Fig. 11, it can be observed that during the peak hours of 0:00–2:00 for residential EV charging and discharging, the highest load is observed in the winter scheme, followed by the autumn scheme. In comparison, there is little difference in load between the spring and summer schemes during this time period. During the secondary peak hours of 18:00–24:00, the ranking of residential EV loads is as follows: spring, winter, autumn, and summer. It is evident that the winter scheme significantly stimulates residential charging loads, effectively encouraging consumption. Therefore, in formulating time-of-use pricing, the elasticity of residential demand relative to price changes plays a crucial role. Winter, with its higher elasticity of residential electricity consumption, has advantages in setting time-of-use pricing: Firstly,

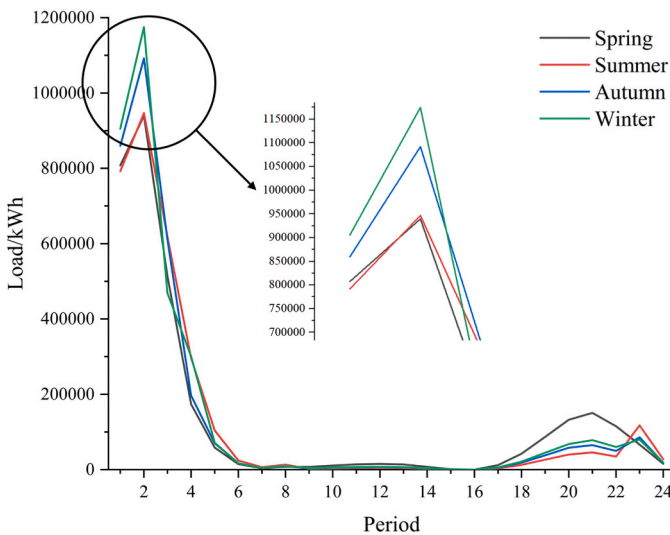


Fig. 11. EV Load Following Implementation of Time-of-Use Pricing Schemes in Typical Seasons.

residents are more responsive to price stimuli, effectively stimulating positive consumption and increasing load demand. Secondly, residents adjust their charging behavior at lower time-of-use pricing levels, further reducing their average charging costs and supporting increased willingness to use EVs. Thirdly, effective positive consumption compensates for grid-side losses. Taking autumn and winter scenarios as examples, winter maintains grid-side benefits comparable to autumn at lower price levels and outperforms autumn schemes in terms of load demand and user benefits.

Furthermore, this section maintains the current residential time-of-use pricing strategy in City F and examines the impact of differences in demand responsiveness during typical seasons on the implementation results of this pricing strategy.

Based on the effectiveness calculations of the current residential time-of-use pricing strategy in City F during typical spring, autumn, and winter seasons, integrated with relevant summer results, a comparison is presented in Table 12.

From Table 12, it is evident that in terms of grid-side benefits, the highest benefit is observed during summer, although the differences among seasons are relatively small, with a maximum variation of 14.89 %. This indicates that the current residential time-of-use pricing strategy in City F has a relatively minor impact on grid profit and loss after implementation in multiple typical seasons. Regarding user-side benefits, summer exhibits the highest average charging cost for users, while the implementation of this time-of-use pricing strategy in other typical seasons results in lower average charging costs for residential EV users. The highest reduction in average charging costs for residents is observed to be 3.85 %. In terms of load demand, while the implementation of this strategy during autumn may lead to a slight decrease in load demand, spring and winter seasons can further stimulate residential EV charging consumption, resulting in an increase in load demand.

8. Conclusion

In this paper, firstly, the price elasticity of EV users is analyzed according to the relationship between EV users' charging and discharging quantity and electricity price; secondly, the Stackelberg model of grid and EV users is constructed, and the benefits of grid and EV users are analyzed by combining the equilibrium tariff scheme and the current tariff scheme of City F; finally, different typical seasonal scenarios are set up to expand the results of the study. The conclusions of the study are as follows:

- (1) The grid and EV users in the Stackelberg game equilibrium can ensure that the benefits of the grid are not impaired, while the total load demand of EV users is only reduced by 8.03 %; in addition, the evening peak load of EV users has been reduced by 27.35 %, and the valley load has been increased by 9.05 %, which verifies the effectiveness of the game model.
- (2) Although Scheme 3 reduces the grid revenue by 74,744.78 CNY, it increases the charging and discharging loads of EV users by 9.83 %. Therefore, when analyzed in terms of cross-subsidies and transmission and distribution fees, the cross-subsidies of Scheme 3 increased by 248 million CNY and 191 million CNY compared to Schemes 1 and 2, respectively, and the transmission and distribution fees of Scheme 3 increased by 87.47 million CNY and 40.34 million CNY compared to

Table 12
Benefits of the Current Residential Time-of-Use Pricing Strategy in Typical Seasons in City F.

Scenario	Grid-Side Benefits (CNY)	User-Side Benefits (CNY/kWh)	Load Demand (kWh)
Spring	−346296.32	0.3794	3399776.08
Summer	−327024.27	0.3873	3380399.16
Autumn	−357840.37	0.3798	3365771.69
Winter	−375724.87	0.3724	3384730.34

Schemes 1 and 2, respectively.

(3) EV users have the highest price elasticity in the winter months, when the grid prefers to reduce tariffs in order to increase load demand: a 10.79 % decrease in peak hour tariffs and a 22.16 % decrease in valley hour tariffs increase demand load by 7.49 %. In contrast, under the current tariff scheme in City F, the changes in grid efficiency and EV users' load demand are less pronounced in all seasons, with maximum differences of 14.89 % and 3.85 %, respectively.

In this paper, the tariff strategy is analyzed using actual charging and discharging load data from EV users in F City, with results compared to the forthcoming tariff policy, ensuring practical applicability. The study innovatively considers seasonal differences in tariff elasticity among EV users. As the number of EVs increases, it is crucial to guide users toward rational charging and discharging through tariff adjustments. The findings are expected to inform policy adjustments in F City, highlighting that the proposed Scheme 3 may lead to significant grid losses during low-demand periods, suggesting that an increase in valley tariffs could be beneficial.

Since this paper focuses on the impact of tariff changes on the grid and EV users, it does not delve deeply into factors affecting EV charging and discharging, such as battery capacity and travel conditions. Future research will consider these factors in depth, further studying the strategy of EV charging and discharging tariffs by incorporating emerging subjects like load aggregators and EV charging station operators.

CRedit authorship contribution statement

Chaofeng Zheng: Writing – review & editing, Validation, Software, Project administration, Methodology. **Zhibin Wei:** Visualization, Validation, Formal analysis. **Lingyan Xu:** Validation, Resources, Investigation, Data curation. **Zhonghai Zheng:** Writing – original draft, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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